

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I Thamaraiselvi S (192511042) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Thamaraiselvi S

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

- (a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
- (b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

- (a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

- (a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
 - (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.
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RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

TASK 1: ARTICLE SUMMARY

(a) Full Citation, Domain and ML Problem

Full Citation

Adua, E., Kolog, E. A., Afrifa-Yamoah, E., Amankwah, B., Obirikorang, C., Anto, E. O., Acheampong, E., Wang, W., & Tetteh, A. Y. (2021). **Predictive model and feature importance for early detection of type II diabetes mellitus.** *Translational Medicine Communications*, 6, 17. <https://doi.org/10.1186/s41231-021-00096-z>.

Domain/Application Area

The article belongs to the **healthcare and medical diagnosis domain**. It focuses specifically on the early detection and prediction of **Type II Diabetes Mellitus (T2DM)**.

Specific ML Problem

The research treats diabetes detection as a **binary classification problem**:

- **0 → Non-T2DM**
- **1 → T2DM**

The authors compare four machine-learning classifiers:

1. Naïve Bayes
2. Support Vector Machine (SVM)
3. K-Nearest Neighbour (KNN)
4. Decision Tree

The study also identifies which patient attributes are most important for predicting T2DM.

(b) Key Objective and Research Gap

Summary – 150–200 Words

The main objective of the article is to develop machine-learning models for the early prediction of Type II Diabetes Mellitus using clinical, anthropometric and biochemical information. The researchers focused on a Ghanaian population and investigated whether machine-learning techniques could identify diabetes-related patterns and important predictive features. The research gap was that conventional prediction methods, particularly traditional regression approaches, may provide limited information about complex relationships among several diabetes risk factors. The authors also noted that previous studies had used different populations, features and machine-learning methods, creating a need for further investigation in an African population. To address this gap, the researchers collected information from 219 T2DM patients and 219 healthy individuals and applied four classification algorithms: Naïve Bayes, KNN, SVM and Decision Tree. The data were cleaned, scaled and analysed using feature-selection techniques. The models were then evaluated using accuracy, precision, recall, F1-score and ROC-AUC. The study also ranked the importance of the input features. The results showed that machine learning could provide useful predictive information for early diabetes detection.

TASK 2: METHODOLOGY ANALYSIS

(a) ML Techniques and Dataset

The researchers used **four machine-learning classification algorithms**:

1. Naïve Bayes

Naïve Bayes is a probabilistic statistical-learning algorithm based on Bayes' theorem. It calculates the probability that a patient belongs to the diabetic or non-diabetic class.

2. Support Vector Machine

SVM identifies a decision boundary that separates the two classes. It is useful when the relationship between features and classes is complex.

3. K-Nearest Neighbour

KNN classifies a new patient according to the classes of nearby observations in the feature space.

4. Decision Tree

The Decision Tree uses feature-based decisions to divide patients into different classes. The study therefore provides a useful comparison of **Statistical Learning, Inductive Learning and Decision Tree approaches**.

Dataset

The dataset contained **438 participants**:

- **219 T2DM patients**
- **219 healthy controls**

The participants were recruited from **Komfo Anokye Teaching Hospital (KATH)** and the local community in Kumasi, Ghana.

Features

The study used 10 predictor variables:

No.	Feature
1	Age
2	BMI
3	Systolic Blood Pressure (SBP)
4	Diastolic Blood Pressure (DBP)
5	HbA1c
6	Fasting Blood Sugar (FBS)
7	Total Cholesterol (TC)
8	Triglycerides (TG)

No.	Feature
9	HDL-C
10	LDL-C

Target variable: T2DM status.

Data Preparation

The researchers:

- Checked the dataset for errors and outliers.
- Found **17 missing values**.
- Used the **Expectation-Maximization (EM)** algorithm to estimate missing values.
- Scaled predictor variables to the range **0–1**.
- Used **Recursive Feature Elimination (RFE)** for feature ranking.
- Used an **80:20 train-test approach** as reported by the authors.

The dataset was already balanced because it contained equal numbers of T2DM and control participants.

(b) Mapping to CO-4 Topics

Technique	CO-4 Topic
Naïve Bayes	Statistical Learning
Decision Tree	Decision Trees
KNN	Inductive Learning
SVM	Statistical/Machine Learning
Feature Selection	Inductive Learning
Classification	Inductive Learning

Methodological Strength

A major strength is that the authors **compared four different ML algorithms** using several evaluation metrics rather than depending on only one model. They also performed feature-importance analysis to determine which variables contributed most to prediction.

Methodological Limitation

One limitation is the **relatively small sample size of 438 participants**. The participants were also obtained from a specific Ghanaian population, so the model may not perform identically on other populations.

The authors themselves acknowledge the small sample size and also mention that factors such as **family history and physical activity** were not included.

TASK 3: RESULTS & CRITICAL EVALUATION

(a) Key Results

The authors reported that all four classifiers achieved performance above their stated 70% threshold for the evaluated measures.

Comparative Results

Model	Accuracy	AUC-ROC
Naïve Bayes	87%	0.87
SVM	84%	0.84
KNN	83%	0.85
Decision Tree	81%	0.81

Naïve Bayes was the best-performing classifier overall. The paper reports that it detected **82% of diabetic test cases**, with **93% precision** for those predicted as diabetic, and an F1-score of approximately **87%** for the diabetic class.

Feature Importance

The three most important features were:

1. **HbA1c**
2. **Total Cholesterol (TC)**
3. **BMI**

The least important features in the reported ranking were:

- Age
- HDL-C
- LDL-C.

(b) Critical Evaluation

The reported results are reasonably realistic because the models achieved **moderate-to-good performance**, rather than unrealistically high values such as 99–100% accuracy.

The AUC values were between **0.81 and 0.87**, indicating good discrimination on this dataset. Naïve Bayes performed best with an AUC of 0.87.

Potential Biases

1. Small Dataset

Only 438 participants were included. A larger dataset could provide more reliable estimates.

2. Population Bias

The participants were from Ghana, so the findings may not generalise directly to populations with different demographic and clinical characteristics.

3. Sampling Bias

The T2DM patients were recruited using purposive sampling, while healthy controls were recruited using convenient sampling.

4. Missing Variables

The authors note that other relevant risk factors, such as family history and physical activity, were not included.

Additional Experiments

The study could be strengthened by:

- Using a larger dataset.
- Using **k-fold cross-validation**.
- Testing the models on an independent external dataset.
- Including family history and physical activity.
- Comparing Logistic Regression and Random Forest.
- Performing subgroup/fairness analysis.
- Reporting confidence intervals.
- Testing model calibration.
- Conducting prospective clinical validation.

(c) Real-World Applicability

The ML models could be used as **clinical decision-support tools** for early diabetes screening.

Possible Applications

- Hospitals
- Primary healthcare centres
- Diabetes screening programmes
- Community health centres
- Telemedicine
- Electronic health-record systems
- Public-health monitoring

Example Workflow

Patient Information



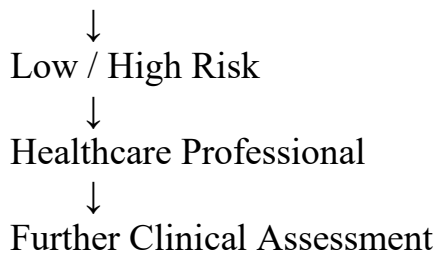
Data Preprocessing



ML Classification Model



Diabetes Risk Prediction



Industry Challenges

Data Availability

Large, high-quality and accurately labelled patient datasets are required.

Data Privacy

Patient information must be securely stored and processed.

Generalisation

A model trained on one population may not perform equally well on another population.

Explainability

Healthcare professionals need understandable reasons for model predictions.

Ethical Issues

Incorrect predictions could affect patient care, so ML systems should be used as **decision-support tools with appropriate clinical oversight**, rather than replacing professional medical judgement.

TASK 4: REFLECTION & LEARNING OUTCOME

(a) Three Key Insights

Insight 1 – Inductive Learning

I learned that machine-learning algorithms can learn general patterns from labelled patient examples and use those patterns to classify new, unseen patients.

CO-4Connection:

This directly relates to **Inductive Learning**, where knowledge is learned from examples and generalised to new cases.

Insight 2 – Decision Trees

The study helped me understand how a Decision Tree can classify patients by repeatedly dividing the data according to predictive features.

CO-4Connection:

This connects to **Decision Trees**, feature selection and classification concepts.

The Decision Tree in this study achieved **81% accuracy** and an AUC of **0.81**.

Insight 3 – Statistical Learning

The most important insight was that a relatively simple statistical-learning method, **Naïve Bayes**, achieved the best overall performance among the four tested classifiers.

It achieved:

- **87% Accuracy**
- **0.87 AUC**
- **87% diabetic-class F1-score**

CO-4Connection:

This directly relates to **Statistical Learning – Naïve Bayes**.

(b) Proposed Extension / Novel Experiment

Proposed Enhancement: Use a Larger Multi-Hospital Dataset

If I were to extend this research, I would use a **larger dataset collected from multiple hospitals and different geographic regions**.

Proposed Method

Multi-Hospital Patient Data



Data Cleaning



Feature Engineering



Feature Selection



Train / Validation / Test



NB + SVM + Decision Tree



Ensemble Model



Diabetes Risk Prediction

Additional Features

The extended dataset could include:

- Family history
- Physical activity
- Dietary habits
- Age
- BMI

- Blood pressure
- HbA1c
- Fasting blood sugar
- Cholesterol

Justification

The original research was conducted using a relatively small Ghanaian cohort. A larger multi-centre dataset would improve the diversity of the training data and allow stronger external validation.

Expected Impact

The proposed enhancement could:

1. Improve generalisation.
2. Reduce population-specific bias.
3. Improve diabetes-risk prediction.
4. Identify important risk factors across different populations.
5. Make the model more suitable for clinical decision-support research.

FINAL CONCLUSION

The article “**Predictive model and feature importance for early detection of type II diabetes mellitus**” is a good choice for this **CO-4 Article Review** because it directly covers **Inductive Learning, Statistical Learning, Decision Trees and feature selection**.

The study compared **Naïve Bayes, SVM, KNN and Decision Tree** using 438 participants. Among them, **Naïve Bayes performed best**, achieving **87% accuracy and 0.87 AUC**, while **HbA1c, Total Cholesterol and BMI** were identified as the three most important predictors.

The research demonstrates that machine learning can support early diabetes-risk prediction, but larger datasets, external validation, additional clinical features and careful ethical evaluation would be necessary before considering real-world clinical deployment.